

MILESTONE 2

NoLoCo Training over AXL

NoLoCo and compression over AXL on CIFAR-10 / ResNet-18

4 / 8 / 16 symmetric workers	11.17M model parameters	5.06× complete-wire reduction
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Report revision 4 | 17 September 2026. This report covers the implementation, measured results, and evidence availability.

Executive summary

- Dromeus implements the data-parallel subset of NoLoCo over AXL, combining local Adam optimization with Nesterov outer momentum and peer-to-peer model updates.
- We completed nine compressed AXL runs at 4, 8, and 16 workers using seeds 17, 29, and 41. Each run trained an 11.17-million-parameter GroupNorm ResNet-18 on CIFAR-10 for 500 outer rounds and 25,000 local Adam steps per worker.
- The independent NCCL reference completed at 4, 8, and 16 workers on matching NVIDIA A10G hardware. A four-worker identity-codec AXL run reached 92.71% mean accuracy, compared with 93.0825% for the matching compressed seed-17 run.
- Top-k sparsification, 8-bit quantization, and error feedback reduced complete update bundles from approximately 89.5 MB to 17.7 MB, about 5.06 times smaller. The compressed runs passed the declared residual bounds and divergence checks.
- Retained evidence includes per-node logs, run manifests, frozen configuration, and validation reports. Large raw captures and some checkpoint binaries were pruned locally after validation; this report identifies which artifacts remain available.

1. Implementation and protocol

M2 implements NoLoCo's data-parallel outer optimizer over asynchronous AXL streams. Training uses pairwise exchanges without a global all-reduce or group-wide training barrier. The official runs use sequential local computation and peer exchange, with overlap disabled. Pipeline parallelism and activation rerouting are outside M2 scope. The optimizer follows the update implemented in Gensyn's official NoLoCo repository. Dromeus expresses the outer gradient as $\mathbf{g} = \mathbf{\phi} - \mathbf{\theta}$, which is mathematically equivalent to the upstream update; compression and error feedback are Dromeus adaptations.

Round lifecycle

1. Seal an even-sized group of 4 to 16 workers using AXL public keys. A seeded perfect matching assigns one peer to each worker in every round; topology alone does not authorize membership.
2. Start from the current slow weights and run 50 local Adam steps with gradient clipping at 1.0. Form the outer gradient as $g = \phi - \theta$, where ϕ denotes slow weights and θ denotes the locally trained fast weights.
3. Build a named two-artifact bundle: `outer_gradient` and `slow_weights`. For outer gradients, add the local error-feedback residual, retain the largest 45% of entries per tensor by magnitude, and quantize values to int8 with bitmap-v2 encoding. Error feedback retains the unsent difference locally: $u = g + e$; $q = \text{decode}(\text{encode}(u))$; $e_{\text{next}} = u - q$. The residual is added to the next outer gradient and is never transmitted. Slow weights use dense int8 quantization without sparsification or error feedback.
4. Exchange readiness and bundle digests with the scheduled peer, then transfer both artifacts in 1 MiB chunks through a four-chunk sliding window. Validate run and round identity, peer authorization, hashes, codec metadata, tensor schema, and finite decoded values before applying an update.
5. Apply the modified Nesterov outer update in FP32 using $\alpha = 0.5$ and $\beta = \gamma = 0.7$. Both peers use decoded local and remote gradients and slow weights; the gradient contribution is shared and the slow-weight correction has opposite signs. Commit the new slow weights, momentum, optimizer state, and residual before advancing. Publish non-blocking CountSketch telemetry separately from training traffic.

Reliability primitives

Primitive	Implementation evidence
Chunk acknowledgments	1 MiB chunks use per-chunk acknowledgments and a four-chunk sliding window; missing chunks can be retried individually.
Retries and deduplication	Retries are bounded at three and duplicate chunks are ignored. A missed pair deadline fails the run explicitly; workers do not rematch or apply stale updates.
Integrity checks	Checksums are checked before transport artifacts or model bundles are applied.
Application flow control	Manifest limits bound message payloads, artifact size, concurrent transfers, in-flight bytes, transfer lifetime, and storage. Received artifacts stream to temporary files before hash validation and atomic finalization.
Peer/update validation	Workers check the round, metadata, artifact, tensor schema, dtype, shape, and finite values.
Durability	Each worker durably records its committed round and local NoLoCo state before advancing. Peers may have different model weights; commit checks validate local integrity rather than requiring equal models. Automatic distributed restart is outside M2 scope.

2. Acceptance criteria mapping

Acceptance criterion	Result	Evidence location
NoLoCo data-parallel training over AXL	COMPLETE	Nine 500-round compressed AXL runs: 4, 8, and 16 workers, seeds 17, 29, and 41. All workers exited successfully.
Benchmark comparison with NCCL at 4, 8, and 16 workers	RUNS COMPLETE	Six retained NCCL references: seeds 17 and 29 at each of 4, 8, and 16 workers. Matched A10G hardware and frozen workload; final accuracy comparisons and recorded timing intervals are reported below.
Compression with documented error-feedback behavior	COMPLETE	45% top-k/int8 outer gradients, dense int8 slow weights, and local error feedback. Approximately 5.06x complete-wire reduction. Residual bounds and divergence checks passed; the W4 identity ablation is retained.
Technical report documenting implementation and benchmarks	DOCUMENTED	Implementation, methodology, benchmark comparisons, M2 charts, limitations, and retained evidence locations are documented here. The report accompanies the v0.2.0 release.
Per-node logs and run manifests for benchmark runs	RETAINED	Per-node logs, AXL run manifests, NCCL run metadata, and checksum records are available in the repository. Appendix B links published logs, manifests, and reports. Access to external checkpoints remains separate.
Release artifacts and supporting evidence	INCLUDED	The v0.2.0 release includes runtime packages, the report, a review bundle, and initial checkpoints. Per-node logs and configuration are linked in Appendix B. Final optimizer checkpoints are separate artifacts.

3. Training benchmark

We trained GroupNorm ResNet-18 (11,173,962 parameters, FP32) on 50,000 CIFAR-10 training examples split into disjoint IID partitions across 4, 8, or 16 workers. Each worker ran 500 outer rounds of 50 local Adam optimizer steps, with batch size 128 and gradient clipping at 1.0. The learning rate warmed from 0.0001 to 0.001 over 1,000 steps, then followed cosine decay to 0.0001 at step 25,000. Evaluation ran every 25 rounds and exact cross-worker weight analysis every 50 rounds.

All workers used g5.xlarge NVIDIA A10G GPUs, with two workers per AWS region. AXL and the independent NCCL reference shared the frozen initialization, data partitions, optimizer settings, and seeded peer schedule. Compressed AXL used seeds 17, 29, and 41 at each scale. NCCL completed seeds 17 and 29 at each of 4, 8, and 16 workers.

The table reports final test accuracy averaged across workers within each run. Delta is AXL minus NCCL in percentage points for matching seeds. A dash indicates that no NCCL run was performed; results are not imputed.

Workers	Seed	AXL accuracy	NCCL accuracy	Delta (pp)
4	17	93.0825%	92.7100%	+0.3725
4	29	92.8500%	92.9550%	-0.1050
4	41	92.3700%	-	-
8	17	93.0788%	93.0913%	-0.0125
8	29	92.6525%	92.8513%	-0.1988
8	41	92.7675%	-	-
16	17	92.9281%	92.6513%	+0.2769
16	29	92.4919%	92.4813%	+0.0106
16	41	92.4344%	-	-

Result interpretation

All six matched comparisons differ by less than 0.38 percentage points in final mean accuracy. These results support comparable convergence on this frozen IID CIFAR-10 workload. NCCL has fewer repetitions than AXL. Regional topology differs between runs, so wall-clock measurements must be reported descriptively without attributing speed differences solely to the transport.

Configuration and provenance

Item	Value
Model / dataset	GroupNorm ResNet-18; 11,173,962 FP32 parameters; CIFAR-10; 50,000 training examples; IID partitions across 4/8/16 workers.
Training	500 rounds x 50 Adam steps; batch 128; clip norm 1.0; 1,000-step warm-up; LR 0.0001 to 0.001 to 0.0001; alpha 0.5, beta = gamma 0.7.
Protocol / codecs	NoLoCo; manifest v3; wire protocol v1; 45% top-k bitmap int8 outer gradients and dense int8 slow weights; 1 MiB chunks; window 4; overlap disabled.
Runtime	Python 3.12.11; PyTorch 2.12.1+cu130; CUDA 13.0; cuDNN 92000; NCCL 2.29.7; NVIDIA driver 595.91.07.
Workers / coverage	g5.xlarge / NVIDIA A10G; two workers per region. Nine compressed AXL runs, six NCCL runs, and one W4 seed-17 identity ablation.

4. Accuracy and loss curves

Figure 1 shows the retained compressed AXL runs for all three seeds at each scale. Each accuracy point is the mean across workers at the recorded evaluation cadence. Loss is the mean of the final local training-batch loss recorded in each outer round, without smoothing; it is not a held-out evaluation-loss curve. No intermediate NCCL curve is inferred from final summaries.

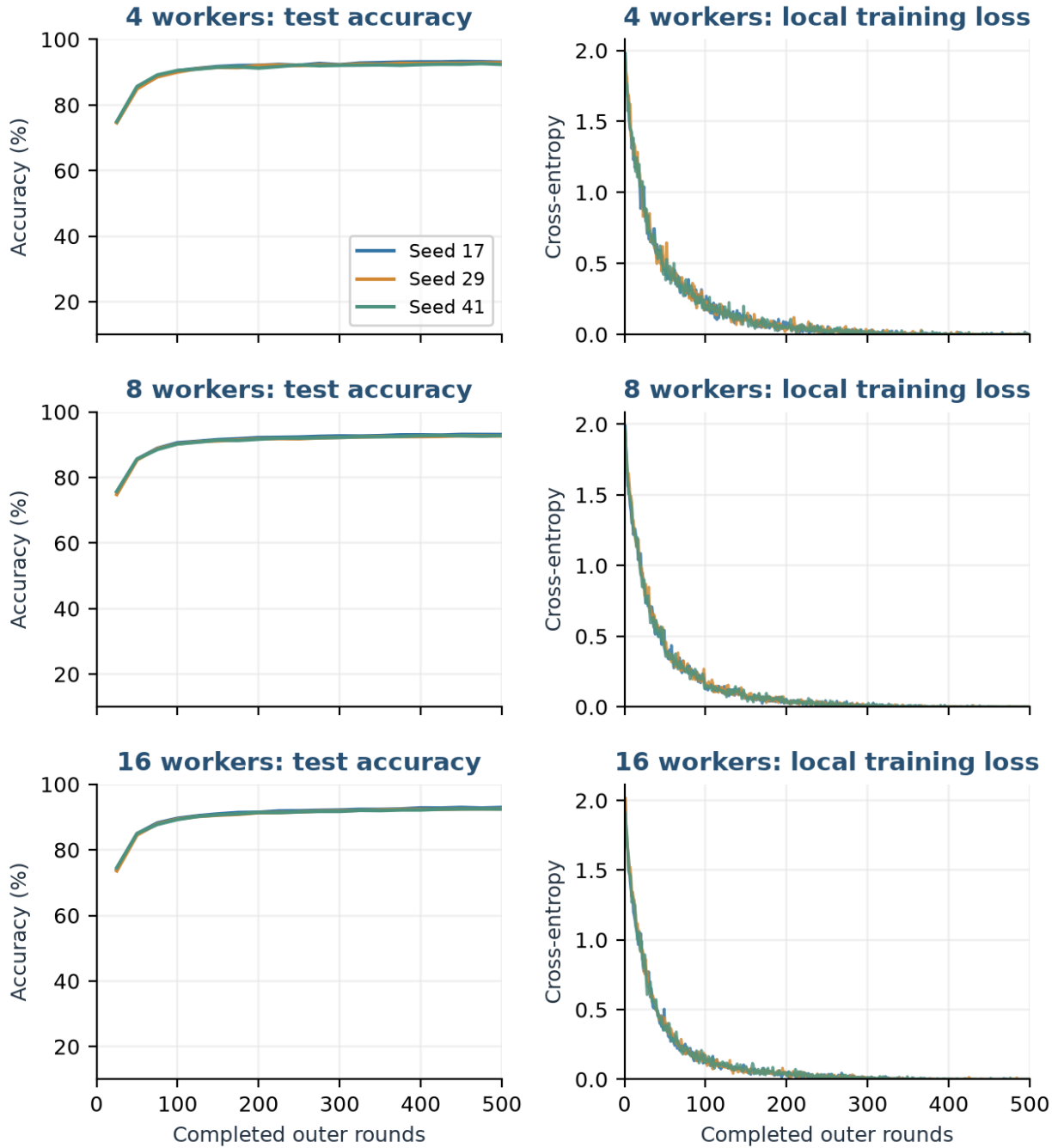


Figure 1. Compressed AXL test accuracy and local training loss over 500 outer rounds. Blue, orange, and green indicate seeds 17, 29, and 41. Each row uses a different worker count.

Across seeds, mean final accuracy was 92.7675% at 4 workers, 92.8329% at 8, and 92.6181% at 16. Accuracy remains close across these scales on the frozen IID workload. Worker-level statistics describe the trained ensemble; workers are not independent experiment repetitions. Appendix A reports worker-level mean, population standard deviation, minimum, and maximum for every accepted run, followed by separate summaries across seeds. No confidence interval or statistical-equivalence claim is made from the limited seed sample.

5. Compression, error feedback, and identity ablation

The outer-gradient codec retains the largest 45% of entries per tensor by magnitude and uses bitmap-v2 support encoding plus int8 values. Slow weights use dense int8 quantization. The outer-gradient residual stores the difference between the compensated input and its decoded representation, then adds that difference to the next outer gradient. Residuals remain local, and slow-weight quantization has no error-feedback state.

Nominal complete-wire accounting in the accepted compression reports is 89,499,868 bytes for an identity bundle and 17,672,081 bytes for a compressed bundle, a 5.06448x reduction. The ratio counts nominal serialized Dromeus transfer-protocol bytes, including codec metadata, envelopes, chunks, and acknowledgments. It excludes retries, telemetry, and underlying AXL/network overhead; it is not a measurement of total physical-network traffic. The identity ablation logs use a narrower application-payload accounting total of about 89,438,360 bytes per node-round; that number is not substituted into the nominal complete-wire ratio.

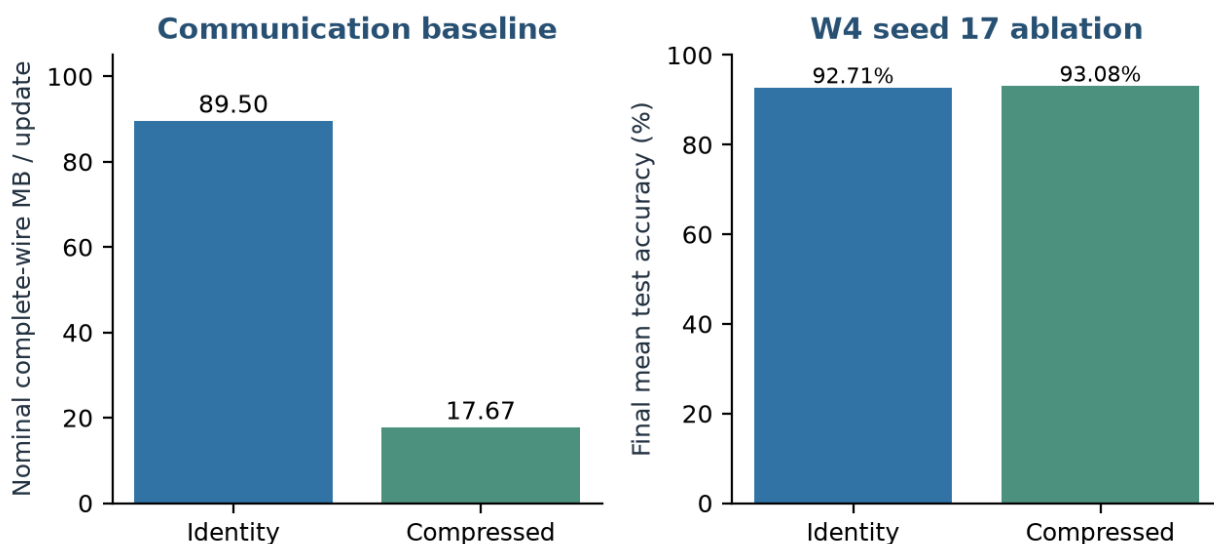


Figure 2. Left: nominal complete-wire bundle sizes from accepted compression accounting, in decimal MB. Right: the W4 seed-17 identity ablation and matching compressed AXL final mean accuracies. The accuracy axis starts at zero.

W4 seed 17	Identity AXL	Compressed AXL
Final mean accuracy	92.7100%	93.0825%
Final rank range	92.56% to 92.84%	93.02% to 93.15%
Local Adam steps / worker	25,000	25,000
Maximum residual L2	0	4.55598

The compressed result was 0.3725 percentage points above identity, within the declared one-point accuracy tolerance. One seed does not establish that compression improves accuracy. The successful identity run used us-east-1 and eu-north-1; the compressed comparison used us-east-1 and eu-central-1. This geographic difference prevents attributing their time difference solely to compression.

Compressed scale	Maximum residual L2	Maximum residual / signal
4 workers	4.63798	0.296900
8 workers	4.70064	0.308071
16 workers	4.91724	0.326298
Frozen bound	6.0	0.35

Each maximum above is taken across all workers and all three seeds at that scale. All nine compressed runs passed both frozen bounds. The identity path recorded exactly zero residual and zero residual-to-signal ratio.

6. Divergence and reliability

NoLoCo trains an ensemble of local models, so success does not require identical worker weights. Exact cross-worker weight standard deviation was measured every 50 outer rounds from temporary FP32 snapshots. The frozen check requires a negative smoothed post-warm-up trend and a final value below the post-warm-up peak; it does not require monotonic decline at every sample.

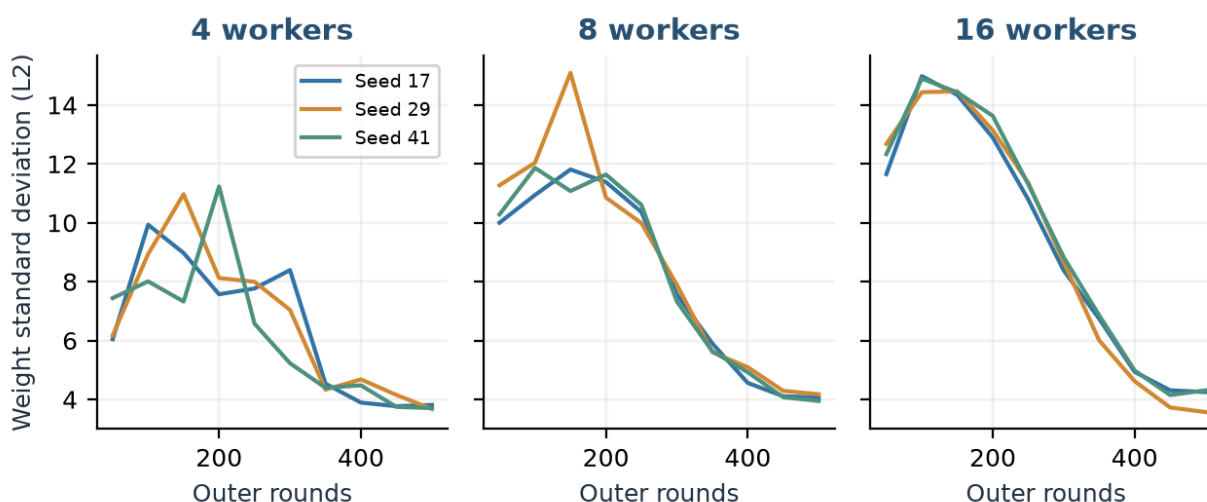


Figure 3. Exact weight standard deviation at the retained analysis cadence for the nine compressed AXL runs. Lines join measured points. Each scale shows all three seeds; lower late-training dispersion is consistent with the decaying learning rate.

All nine compressed runs passed the recorded trend check. The W4 identity ablation also passed, with slope -0.000398755 per inner step and final weight standard deviation L2 of 1.28452. Final committed state and evidence logs recorded 500 rounds for each accepted run.

Strict reconciliation covered 42,000 compressed AXL node-rounds: 6,000 at W4, 12,000 at W8, and 24,000 at W16. The identity ablation added 2,000 node-rounds. Checks distinguish algorithm-update payloads, telemetry, and retry counters. The identity rerun recorded one metric-level retry even though transport events labeled zero retries; these counters are not interchangeable.

The first full identity attempt failed at round 360 with a pair deadline timeout and was excluded. The accepted rerun completed all 500 rounds. Earlier W4 AXL evidence was replaced by fresh runs with separately captured AXL-process logs; W8 NCCL seed 17 was rerun to capture final model weights. These are disclosed evidence replacements, not extra independent seeds. Accepted results are conditional on the completed runs, and do not measure an overall infrastructure success rate.

7. Timing and deployment conditions

Figure 4 separates the per-node timing fields emitted by Dromeus: local compute, peer wait, transfer, outer update, and evaluation. Values are means over 500 rounds and all workers within each run. Their sum is the recorded node-round duration, not the elapsed wall time of a synchronized global training loop.

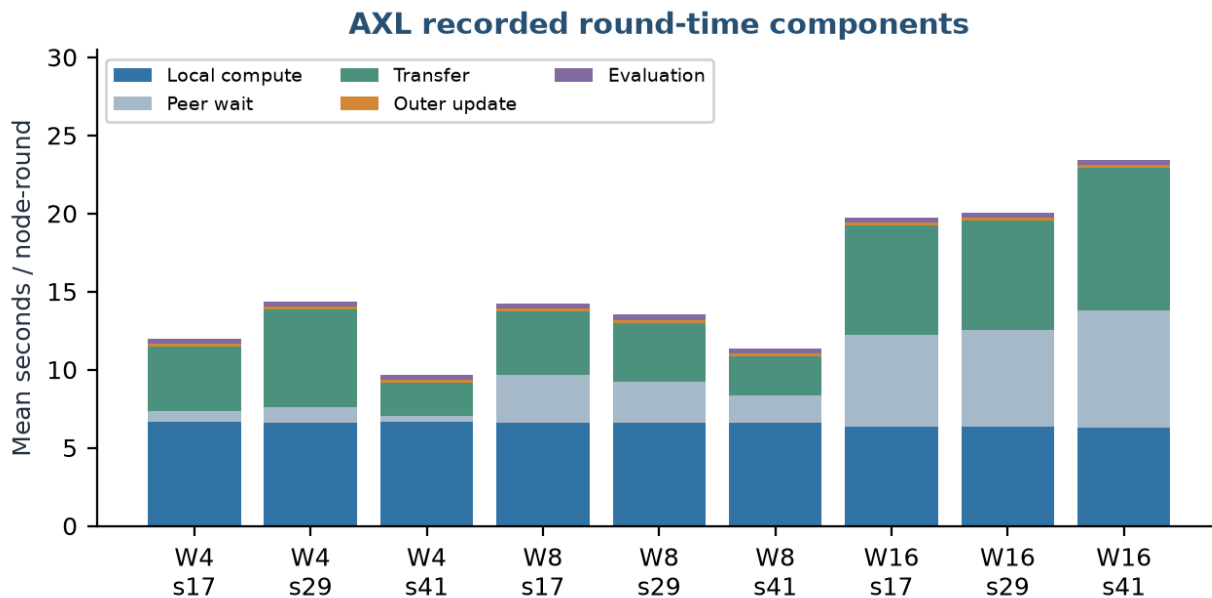


Figure 4. Mean recorded AXL timing components by scale and seed. These include application overhead and the observed regional network conditions. Overlap was disabled in every official run.

Scale / seed	AXL lifecycle (min)	NCCL recorded interval (min)
4 / 17	117.78	34.37
4 / 29	137.91	58.40
4 / 41	98.30	Not run
8 / 17	135.72	53.02
8 / 29	129.85	44.60
8 / 41	112.17	Not run
16 / 17	184.35	62.22
16 / 29	186.36	74.58
16 / 41	211.09	Not run

AXL lifecycle is measured from the earliest node_start to the latest node_complete event, including formation, evaluation, and shutdown work. NCCL intervals use the retained execution reports or the earliest start/latest end markers for W8 seed 17. The timing boundaries and network fabrics differ, so these are descriptive durations, not a normalized speedup benchmark. The per-node AXL timing sums are reported separately in Figure 4. GPU provisioning and capacity waits are excluded.

All workers used NVIDIA A10G GPUs on g5.xlarge instances, two per AWS region. The reference used NCCL Socket transport over private VPC peering. AXL used separately managed routing processes and the local HTTP bridge. Example regions include us-east-1 and eu-central-1 for W4 compressed seed 17, us-east-1 and eu-north-1 for W4 identity, and eight regions for W16 NCCL seed 17: us-east-1, us-west-2, us-east-2, eu-central-1, eu-west-2, eu-west-3, eu-north-1, and sa-east-1. Per-run deployment records retain the exact topology.

Mean recorded AXL transfer time ranged from 2.14 to 6.26 seconds per node-round at W4, 2.51 to 4.01 seconds at W8, and 6.98 to 9.13 seconds at W16. These observations support the communication analysis but do not isolate region, routing, or transport overhead experimentally. Task-owned compute and network resources were cleaned after evidence capture.

8. Correctness, provenance, and reproduction

The NoLoCo outer optimizer follows the pinned gensyn-ai/noloco executable semantics from `sparse_optimizer_c.py` at commit `a1b4a425bdc4050a356cf9f4bae7c383419703ab`. The descent-oriented convention $g = \phi - \theta$ is algebraically equivalent to the source update; the discrepancy with the paper's printed sign is documented. The independent NCCL adapter shares the frozen workload and scheduler but implements its outer optimizer and exchange outside the production Dromeus runtime.

Correctness evidence includes an independently generated upstream fixture, hand-computed traces, and a four-node, two-round deterministic AXL/NCCL trajectory comparison. All 12 initialization/post-outer snapshots matched with zero absolute and relative error at $\text{atol} = \text{rtol} = 1\text{e-}6$. This bounded parity test is distinct from the 500-round accuracy comparisons. Multi-chunk tests covered loss, duplicate delivery, reordering, corruption, and interruption. Gloo runs are development checks and are not the NCCL acceptance reference.

Pinned input	Identity
Dromeus benchmark runtime	411daee79fffb4c6dbeab9d65dfe0280d0492a61
AXL commit	628e28ace077f26dfe8d0259009b357216a9d8d4
Frozen experiment file SHA-256	c588c5b389bf0b506723786b351f4bd816608230b29628bc9596021fa57ea5cd
Container digest	sha256:1b40d3774dc864f1f4450720f91243f2e379de8ec7d12805d120ec1ef1419a3c
Runtime versions	Python 3.12.11; PyTorch 2.12.1+cu130; CUDA 13.0; NCCL 2.29.7; driver 595.91.07

Reproduction uses the shared `experiment.yaml` and its hash-bound initial checkpoints and pilot record, then selects world size and seed through the official benchmark runner. Addresses and machine-local paths must be provisioned anew without changing the frozen experiment. A current development checkout is not automatically equivalent to the recorded benchmark runtime. Check package-relative checksums and run manifests before analysis.

9. Evidence availability and limitations

Evidence	Retained location under <code>benchmarks/results/m2/</code>
Compressed AXL runs	<code>benchmark-results/{4,8,16}-nodes/seed-{17,29,41}/</code>
NCCL reference	<code>nccl-reference/4-nodes/seed-{17,29}/</code> ; <code>8-nodes/seed-{17,29}/</code> ; <code>16-nodes/seed-{17,29}/</code>
Identity ablation	<code>identity-ablation/4-nodes/</code>
Frozen configuration	<code>experiment-config/experiment.yaml</code> and initial checkpoint files

Evidence	Retained location under benchmarks/results/m2/
Report figures and source hashes	report/charts/ and report/chart-data.json

The retained packages include per-node logs, manifests or reference run metadata, summaries, validation reports, deployment provenance, and checksums. AXL final-checkpoint JSON records identify versioned S3 objects that require authorized access; those objects are separate from the public download. Earlier NCCL final model-weight binaries were validated and later pruned locally; their hashes remain. The W16 seed-29 rerun retains all 16 verified final model-weight files locally outside Git. Its rank-9 torchrun exit-barrier warning occurred after the final checkpoint and summary; all 16 launchers exited 0. The prior host-maintenance failure is excluded; its separate raw capture is no longer available locally. The identity package retains final model-weight files, but its full-state checkpoint binaries and raw staging capture were pruned. A recorded hash is an integrity reference, not a downloadable copy.

Large local archives and temporary analysis tensors were removed after validation. The document therefore does not promise a self-contained reproduction archive or publicly downloadable checkpoint set. Appendix B provides pinned links to the repository evidence. The source repository is <https://github.com/devesh1011/dromeus>; the upstream reference is <https://github.com/gensyn-ai/noloco>. The v0.2.0 release provides the runtime distributions and review artifacts. Appendix B links the published review package and records external or missing artifacts.

The workload uses IID CIFAR-10 partitions and an approximately 11-million-parameter vision model. These results do not establish non-IID convergence, large-language-model quality, pipeline-parallel operation, Byzantine robustness, secure aggregation, or automatic recovery after membership failure. Fixed membership and bounded pair progress remain assumptions. Unequal regional topologies and limited NCCL repetitions constrain statistical and performance conclusions.

10. Milestone outcome

The retained evidence demonstrates NoLoCo data-parallel training over AXL at 4, 8, and 16 workers, an independent NCCL comparison at those scales, and a working compression layer with documented error feedback. Nominal complete-wire compression is approximately 5.06x, with all recorded compressed residual bounds and divergence checks passing. The W4 identity ablation provides the uncompressed quality control.

This report records the implementation and measured benchmark outcomes. The v0.2.0 release provides runtime packages, this report, and compact evidence, with per-node logs linked in Appendix B. No additional training repetitions are planned for this submission.

Appendix A. Benchmark statistics

Each row summarizes final test accuracy across workers within one accepted run. SD is population standard deviation across workers, in percentage points (pp). AXL denotes the final 45% compressed configuration; identity is listed separately. Workers are not independent experiment repetitions.

Method	Workers	Seed	Mean (%)	SD (pp)	Min (%)	Max (%)
AXL	4	17	93.0825	0.0507	93.0200	93.1500
AXL	4	29	92.8500	0.0883	92.7500	92.9900
AXL	4	41	92.3700	0.0442	92.3000	92.4200
AXL	8	17	93.0788	0.0836	92.9600	93.2200
AXL	8	29	92.6525	0.0845	92.4700	92.7500
AXL	8	41	92.7675	0.0672	92.7000	92.9000
AXL	16	17	92.9281	0.0446	92.8400	93.0400
AXL	16	29	92.4919	0.0573	92.3500	92.5800
AXL	16	41	92.4344	0.1541	92.0800	92.6900
NCCL	4	17	92.7100	0.1082	92.5600	92.8400
NCCL	4	29	92.9550	0.0512	92.9100	93.0400
NCCL	8	17	93.0913	0.0909	92.9300	93.2000
NCCL	8	29	92.8513	0.1043	92.6900	93.0300
NCCL	16	17	92.6513	0.1679	92.3400	92.9300
NCCL	16	29	92.4813	0.1195	92.2900	92.6800
Identity AXL	4	17	92.7100	0.1082	92.5600	92.8400

Source: retained per-run acceptance reports; W8 NCCL seed 17 uses its accepted remediation summary. Values are rounded for display. No missing NCCL seed-41 result is imputed.

Summary across completed seeds

Each seed contributes one worker-mean accuracy with equal weight. SD below is the sample standard deviation across seed means ($n - 1$ denominator), not worker dispersion. Min and max are the smallest and largest seed means. AXL uses seeds 17, 29, and 41; NCCL uses 17 and 29. These unequal seed sets are descriptive summaries; paired comparisons remain in Section 3.

Method	Workers	Seeds	Mean (%)	SD (pp)	Min (%)	Max (%)
AXL	4	3	92.7675	0.3633	92.3700	93.0825
NCCL	4	2	92.8325	0.1732	92.7100	92.9550
AXL	8	3	92.8329	0.2205	92.6525	93.0788
NCCL	8	2	92.9713	0.1697	92.8513	93.0913
AXL	16	3	92.6181	0.2700	92.4344	92.9281
NCCL	16	2	92.5663	0.1202	92.4813	92.6513

The identity ablation has one seed, so no across-seed standard deviation is estimated. No confidence interval or statistical-equivalence claim is made.

Historical matched transfer timing

An earlier matched four-worker pilot used the same A10G instances and AXL topology for identity and 40% bitmap-int8 compression over 20 rounds. Mean transfer time fell 67.58%. That historical pilot used 40% retention and a shorter schedule; it does not establish a controlled transfer-time reduction for the final 45% configuration.

Pilot date: 23 August 2026; four workers; 20 outer rounds and 1,000 inner steps; identity and 40% compression reused the same instances, identities, topology, initialization, partitions, seeds, schedule, and CUDA environment. This is historical diagnostic evidence, separate from the final 45% official matrix.

Appendix B. Reviewer evidence index

Selected report links below remain pinned to repository commit 1734c5d368af68130dd116b6b7d1667442680227. The six acceptance reports and frozen experiment match the retained files. The full current public package is linked after the table, verified on 17 September 2026.

Evidence	Verified public link	Coverage
Frozen experiment	experiment.yaml	Public; file matches the SHA-256 in Section 8.
W8 compressed AXL	Seed 17 Seed 29 Seed 41	All three acceptance reports match the retained local versions.
W16 compressed AXL	Seed 17 Seed 29 Seed 41	All three acceptance reports match the retained local versions.
Source repository	Pinned public source tree	Public source/evidence snapshot; the benchmark runtime remains the separate commit pinned in Section 8.

Published evidence and external artifacts

The current public package includes the latest W4 compressed reruns, all six NCCL references, the W4 identity ablation, chart sources, and retained per-node logs and manifests. Use the current evidence index below for the accepted packages. Earlier W4 records remain historical.

[Current M2 evidence index](#) | [Artifact availability and retention](#)

The repository includes retained per-node logs, manifests, and NCCL run metadata. The release includes initial model weights. Final model and optimizer checkpoint files are external artifacts with separate access requirements. This is not a self-contained training reproduction archive.

W16 NCCL seed-29 final model weights and identity final model weights remain local. Earlier pruned NCCL binaries remain unavailable locally; recorded hashes do not restore those files. See Section 9 for retention limits.

Release artifacts: v0.2.0 contains the runtime wheel and source distribution, this report, the M2 review bundle, and frozen initial checkpoints. Checksums accompany the downloadable release assets.